HKU MAIB7002 | GROUP Q | FRESH RETAIL

FRESH FOOD. SMARTER DISCOUNTS.

Should we discount it tomorrow?



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Can a better forecast guide tomorrow's discount?

A Dingdong category manager must balance unsold perishables against value lost to markdowns. We first predict sales, then test what happens when the forecast chooses the price.

1 Choose histories with price changes

We chose FreshRetailNet for its daily store-product sales, discounts and stockout records. We needed products with regular sales and repeated discount changes.

50,000 public histories → **16,684** pass the filters → **312** in five stores

Keep histories with 15-85% discounted days (rate < 0.95), 6+ discount-state changes and <20% zero-sales days. Select the five stores with the most qualifying histories.

122 products, one city; 25,896 training/validation rows.

2 Test our assumptions first

We began with kNN: predict from similar past days. Adding weather looked useful, but validation showed that fewer inputs worked better.

Mean absolute error (MAE)

With weather	0.3267
Weather removed	**0.3120**

kNN: k=25, equal weights; five validation weeks.

Decision: keep 10 inputs covering sales history, stockouts, discounts and calendar information.

3 Build from a simple baseline to a combined forecast

Predict next-day normalized sales; select settings using five time-ordered validation weeks.

Random Forest
300 parallel trees
Average their predictions

CatBoost
500 sequential trees
Each corrects prior errors

→ **0.5 RF + 0.5 CatBoost**

Averaging helped slightly. We chose 50/50 for simplicity on a flat validation curve; CatBoost alone is almost tied on the final week.

Final-week MAE (lower is better)

7-day average	0.3142
kNN	0.3040
Random Forest	0.2923
CatBoost	0.2896
50/50 ensemble	**0.2892**

7.9% lower MAE

vs. the 7-day average | RMSE 0.4593

Validation: 22 May - 25 Jun 2024. Final evaluation: 26 Jun - 2 Jul, 2,184 cases (a reused benchmark).

4 Then the model wants to discount

We varied tomorrow's price rate over eight candidates and maximized predicted sales x rate. Even with historical support, it chose discounts almost every day.

98.3% discount advised

Model + historical support

Require 3+ training days within 0.025 of a candidate rate. This alone did not make recommendations selective.

5 Put stock rules before price

We therefore tested rules: stockout today means full price; no stockout in a week requires a discount. Otherwise, the model compares eligible rates.

55.0% discount advised

Model + support + stock rules

The rules change the recommendation rate. They assume a stock policy; stockout records do not reveal inventory age.

Same 2,184 cases. Zero unit cost assumed. These are recommendation rates, not measured business improvements.

6 One product: 5% off

Store 18 / Product 11 / 26 Jun 2024

Highest predicted value among eligible rates.

Full price	5% off	20% off
1.121	**1.198**	**1.136**

Predicted value = sales x rate (5% off means a rate of 0.95).

Next: test the pricing decision in a real store.

Sales are normalized and stock-limited; discounts are observational. No measured revenue or waste gain.
Five stores, retrospective selection and a reused evaluation week limit generalization.

FreshRetailNet-50K / Dingdong / CC BY 4.0. Final notebook: pricing-ml-ap @ a7e26dd. Add inventory/cost data before a pilot.

CODE + NOTEBOOK